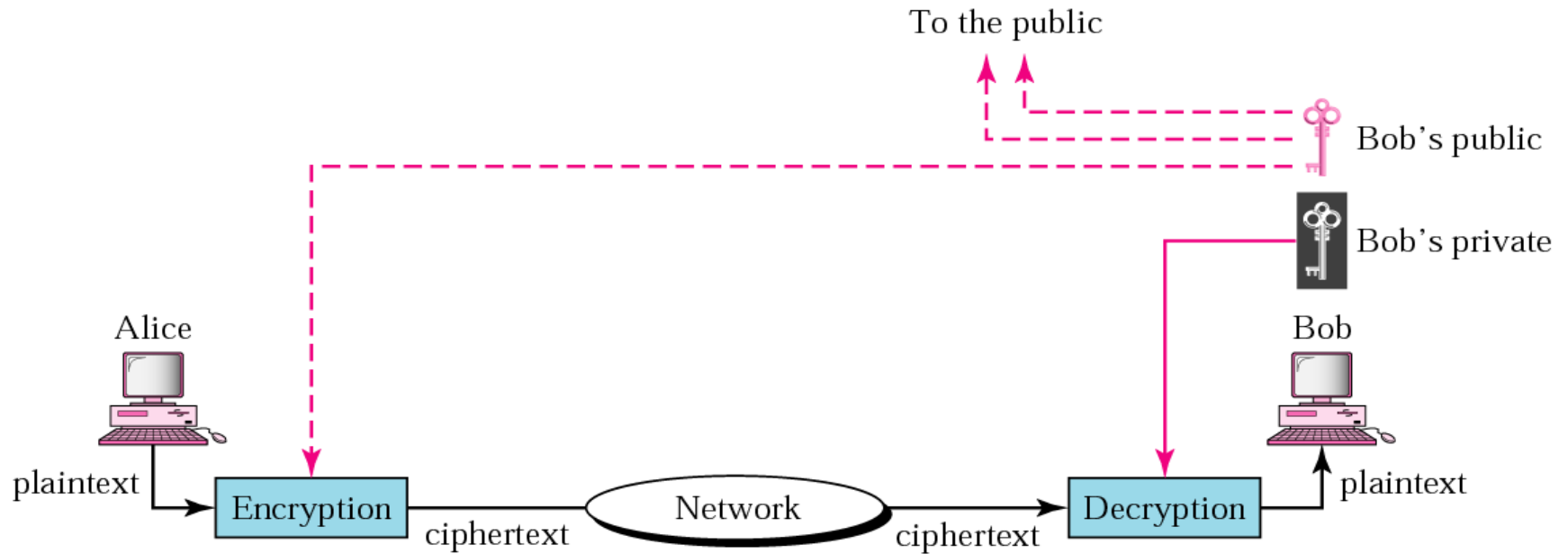


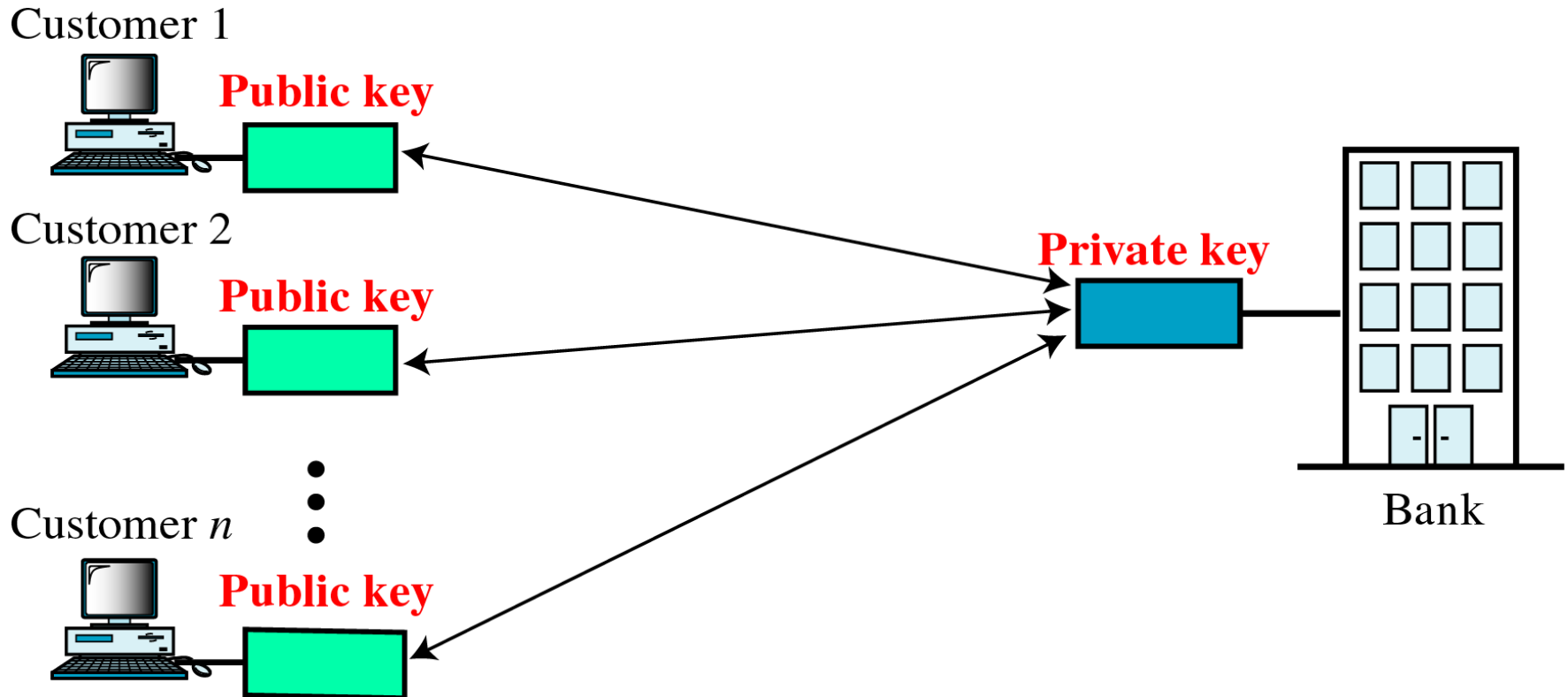
Public Key Cryptography

Background



Public-key algorithms are more efficient for short messages.

Background



Symmetric vs. Public-Key Encryption

Symmetric Encryption (Conventional)	Public-Key Encryption (Public)
The same algorithm with the same key is used for encryption / decryption	One algorithm is used for encryption and decryption with a pair of keys, one for encryption and one for decryption
The sender and receiver must share the algorithm (name) and the secret key (securely)	The sender and receiver must know the algorithm and each have one of the matched pair of keys
It must be impossible or at least impractical to decipher a message if no other information is available	It must be impossible or at least impractical to decipher a message if no other information is available,
Knowledge of the algorithm plus samples of ciphertext must be insufficient to determine the key	Knowledge of the algorithm plus one of the keys plus samples of ciphertext must be insufficient to determine the other key

Public-Key Applications

- **Can classify uses into 3 categories:**
 - encryption/decryption (provide secrecy),
 - digital signatures (message authentication),
 - key exchange (of session or shared keys)
- **Some algorithms are suitable for all uses, others are specific to one**

Algorithm	Encryption/Decryption	Digital Signature	Key Exchange
RSA	Yes	Yes	Yes
Elliptic Curve	Yes	Yes	Yes
Diffie-Hellman	No	No	Yes
DSS	No	Yes	No

Public-Key Requirements

- **Public-Key algorithms rely on two keys where:**
 - it is computationally infeasible to find decryption key knowing only the algorithm & the encryption key,
 - it is computationally easy to en/decrypt messages when the relevant (encryption/decryption) key is known,
 - either of the two related keys can be used for encryption, with the other used for decryption
- **Only a few algorithms have satisfied the above requirements**

Security of Public-Key Schemes

- Like secret key schemes, brute force exhaustive search attack is always theoretically possible,
- But keys used are too large (>512 bits),
- Thus the size of the key usually determines the level of security,
- Hence, in some cases, it is required to use very large numbers,
- Consequently, it is slow compared to private key schemes and recommended only for short messages

The RSA Public Methodology

- By Rivest, Shamir & Adleman of MIT in 1977,
- Best known & widely used public-key scheme,
- Based on a solid mathematical background over integers modulo operations,
- Uses large integers (e.g. 1024 bits),
- Security comes from the cost of factoring large numbers

RSA Encryption/Decryption

- **To encrypt a message M the sender:**
 - obtains public key of recipient $PU=\{e, n\}$
 - computes: $C = M^e \bmod n$, where $0 \leq M < n$
- **To decrypt the ciphertext C the owner:**
 - uses their private key $PR=\{d, n\}$
 - computes: $M = C^d \bmod n$
- **Note that the message M must be smaller than the modulus n (use blocks if needed)**

RSA Key Setup

- **Users generate a public/private key pair by:**
 - selecting two large primes at random: p, q
 - compute $n = p \cdot q$
 - note $\phi(n) = (p-1)(q-1)$
 - selecting at random the encryption key e
 - where $1 < e < \phi(n)$, $\gcd(e, \phi(n)) = 1$
 - solve following equation to find decryption key d
 - $e \cdot d \bmod \phi(n) = 1$ and $0 \leq d \leq n$
 - publish their public encryption key: $PU = \{e, n\}$
 - keep secret private decryption key: $PR = \{d, n\}$

RSA Key Setup Example

1. **Select primes: $p=17$ & $q=11$**
2. **Calculate $n = p \times q = 17 \times 11 = 187$**
3. **Calculate $\phi(n) = (p-1)(q-1) = 16 \times 10 = 160$**
4. **Select e : $\gcd(e, 160) = 1$; choose $e=7$**
5. **Determine d : $d \cdot e \bmod 160 = 1, d < 160$
choose $d=23$ since $23 \times 7 = 161 \bmod 160 = 1$**
6. **Publish public key $PU = \{7, 187\}$**
7. **Keep secret private key $PR = \{23, 187\}$**
- 8.

RSA En/Decryption Example

➤ **A sample RSA encryption/decryption is:**

➤ **given message $M = 88$ ($88 < 187$)**

➤ **encryption:**

$$C = 88^7 \bmod 187 = 11$$

➤ **decryption:**

$$M = 11^{23} \bmod 187 = 88$$

Real RSA Example

Let us give a realistic example. We randomly chose an integer of 512 bits. The integer p is a 159-digit number.

p = 96130345313583504574191581280615427909309845594996215822583150879647940
45505647063849125716018034750312098666606492420191808780667421096063354
219926661209

The integer q is a 160-digit number.

q = 12060191957231446918276794204450896001555925054637033936061798321731482
14848376465921538945320917522527322683010712069560460251388714552496900
0359660045617

Real RSA Example (Continue ...)

We calculate n . It has 309 digits:

```
n = 11593504173967614968892509864615887523771457375454144775485526137614788
54083263508172768788159683251684688493006254857641112501624145523391829
27162507656772727460097082714127730434960500556347274566628060099924037
10299142447229221577279853172703383938133469268413732762200096667667183
1831088373420823444370953
```

We calculate Φ . It has 309 digits:

```
φ = 11593504173967614968892509864615887523771457375454144775485526137614788
54083263508172768788159683251684688493006254857641112501624145523391829
27162507656751054233608492916752034482627988117554787657013923444405716
98958172819609822636107546721186461217135910735864061400888517026537727
7264467341066243857664128
```

Real RSA Example (Continue ...)

We choose $e = 35,535$. We then find d .

$e = 35535$

$d = 58008302860037763936093661289677917594669062089650962180422866111380593852$
82235873170628691003002171085904433840217072986908760061153062025249598844
48047568240966247081485817130463240644077704833134010850947385295645071936
77406119732655742423721761767462077637164207600337085333288532144708859551
36670294831

Alice wants to send the message “THIS IS A TEST” which can be changed to a numeric value by using the 00–26 encoding scheme (26 is the space character).

$P = 1907081826081826002619041819$

Real RSA Example (Continue ...)

The ciphertext calculated by Alice is $C = P^e$, which is.

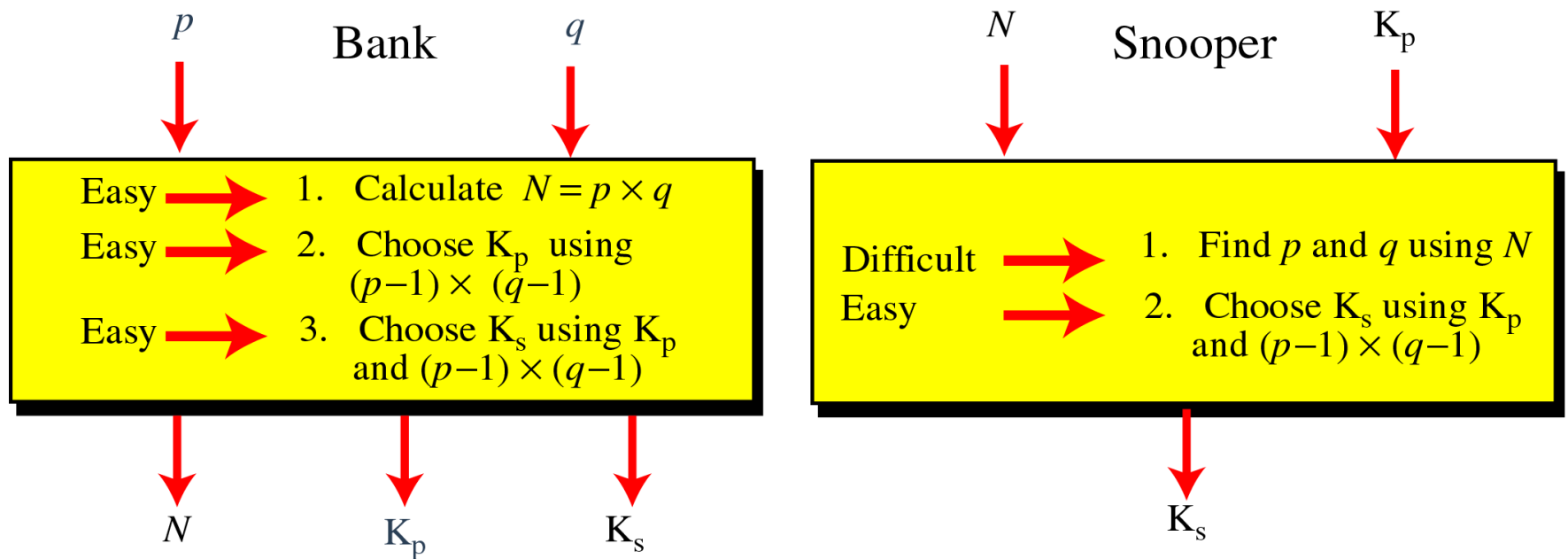
```
C = 4753091236462268272063655506105451809423717960704917165232392430544529
6061319932856661784341835911415119741125200568297979457173603610127821
8847892741566090480023507190715277185914975188465888632101148354103361
6578984679683867637337657774656250792805211481418440481418443081277305
9004692874248559166462108656
```

Bob can recover the plaintext from the ciphertext by using $P = C^d$, which is

```
P = 1907081826081826002619041819
```

*The recovered plaintext is **THIS IS A TEST** after decoding.*

Security of the RSA Methodology



Diffie–Hellman Key Exchange

- **Diffie–Hellman key exchange establishes a shared secret between two parties that can be used for secret communication for exchanging data over a public network,**
- **An analogy illustrates the concept of public key exchange by using colors instead of very large numbers is given by the next slide:**

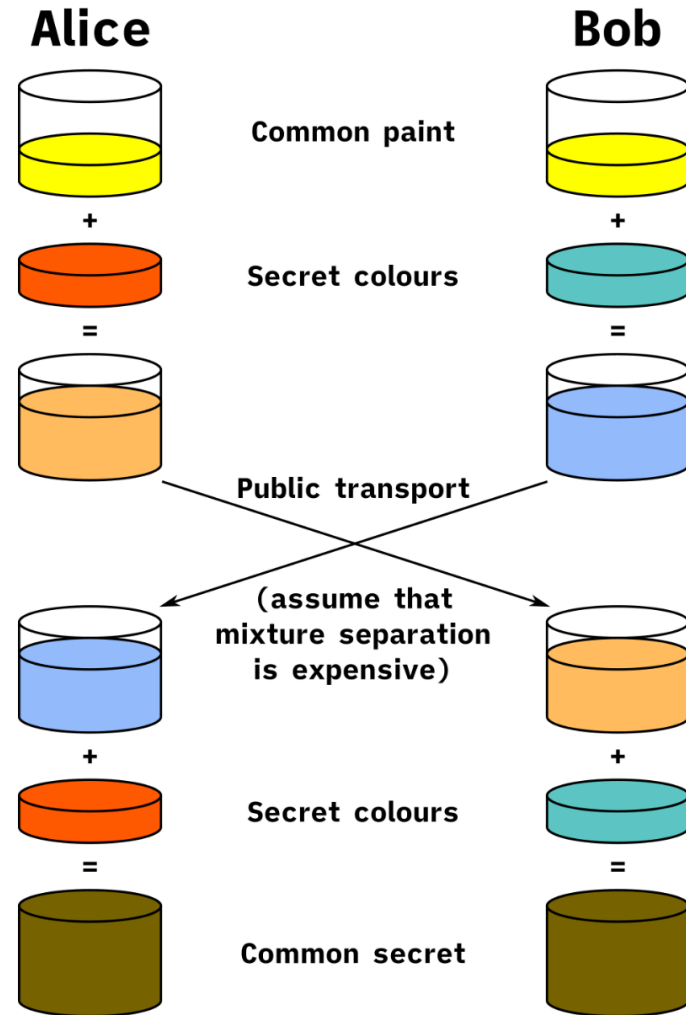
Diffie-Hellman

➤ The process begins by having the two parties, Alice and Bob, publicly agree on an arbitrary starting color that does not need to be kept secret. In this example, the color is yellow,

➤ Each person also selects a secret color that they keep to themselves, in this example, red and cyan,

➤ The crucial part of the process is that Alice and Bob each mix their own secret color together with their mutually shared color, resulting in orange-tan and light-blue mixtures respectively, and then publicly exchange the two mixed colors,

➤ Finally, each of them mixes the color they received from the partner with their own private color, the result is a final color mixture (yellow-brown in this case) that is identical to their partner's final color mixture.



The Diffie–Hellman Algorithm

- The Diffie-Hellman (DH) algorithm is a cryptographic method allowing two parties (Alice and Bob) to securely establish a shared secret key over an insecure channel, like the internet, without ever sending the key itself, using public parameters and modular arithmetic (discrete logarithms),
- This shared secret, generated independently by both sides, then becomes the key for symmetric encryption (like AES) for their private communication,
- Its security relies on the difficulty of solving the discrete logarithm problem, making it vulnerable to man-in-the-middle attacks

How it works

- **Agree on Public Numbers:** Alice and Bob publicly agree on a large prime number (p) and a base (g) (Diffie-Hellman parameters),
- **Generate Private Secrets:** Alice chooses a secret private number (a), Bob chooses a secret private number (b),
- **Exchange Public Values:** Alice calculates ($A = g^a \bmod p$) and sends to Bob, Bob calculates ($B = g^b \bmod p$) and sends to Alice,
- **Compute Shared Secret:** Alice computes ($S = B^a \bmod p$), Bob computes ($S = A^b \bmod p$),
- **Result:** Both Alice and Bob arrive at the identical shared secret (S), which an eavesdropper (Eve) cannot easily determine from the public (p, g, A, B) without solving a complicated discrete logarithm problem,
- **Key Aspects Purpose:** Key exchange, not encryption itself,
- **Significance:** One of the first public-key cryptography methods, enabling secure communication over insecure channels

How it works (Continue)

